ADVANCED LINEAR ALGEBRA
HOMEWORK 4

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P.2.9 Let $\mathbf{u} = [1 \quad 0 \quad 1]^\top$, $\mathbf{v}_1 = [0 \quad 1+i \quad 1-i]^\top$, $\mathbf{v}_2 = [2i \quad 1 \quad 0]^\top$, and $\mathbf{v}_3 = [2 \quad -1 \quad 1]^\top$. Let $\beta = \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. Show that β is a basis for $\mathbb{C}^3$ and compute $[\mathbf{v}]_\beta$.

Proof. Sol □

P.2.11 Compute rank(A), in which

$$A = \begin{bmatrix} 0 & 3 & 1 & 5 & 0 & 5 & 1 & 4 & 4 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 & 3 & 2 & 1 & 5 \\ 1 & 4 & 0 & 1 & 5 & 3 & 3 & 0 & 0 & 0 \\ 2 & 4 & 5 & 0 & 2 & 0 & 2 & 0 & 4 & 3 \end{bmatrix} \in \mathbf{M}_{4\times 10}.$$

Hint. Find four columns whose span includes $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4$.

Solution. Sol □

P.2.16 Let $\beta = \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ be a basis for a non-zero $\mathbb{F}$-vector space $\mathcal{V}$. (a) If any vector in $\mathcal{V}$ is appended to β, explain why the resulting list still spans $\mathcal{V}$ but is not linearly independent. (b) If any vector in β is omitted, explain why the resulting list will still be linearly independent but no longer spans $\mathcal{V}$.

Solution. Sol □

P.2.18 Let $\mathcal{V}$ be an n-dimensional $\mathbb{F}$-vector space with $n \geq 2$ and let $\beta = \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ be a list of vectors in $\mathcal{V}$ with $1 \leq r < n$. Show that β does not span $\mathcal{V}$.

Proof. Sol □

P.2.41 What is the rank of

$$A = \begin{bmatrix} 1 & n+1 & 2n+1 & \cdots & (n-1)(n+1) \\ 2 & n+2 & 2n+2 & \cdots & (n-1)(n+2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & 2n & 3n & \cdots & n^2 \end{bmatrix}?$$

Find a basis for col(A).

Solution. Sol □

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P.3.2 Let

$$A = \left[\begin{array}{c|cc} -1 & 2 & 3 \\ \hline 2 & -3 & 1 \\ 3 & 1 & -2 \end{array}\right] = \left[\begin{array}{cc|c} -1 & 2 & 3 \\ \hline 2 & -3 & 1 \\ 3 & 1 & -2 \end{array}\right] \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 3 & -2 \\ 3 & -2 & 1 \\ -2 & 1 & 3 \end{bmatrix}$$

(a) Compute AB in two different ways by partitioning B conformally with each presentation of A and then by preforming block matrix multiplication. (b) Compute AB in standard manner, and verify that all three answers agree

Solution. Sol $\square$

P.3.3 Let

$$A = \left[\begin{array}{c|cc|cc} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ \hline a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ \hline a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \end{array}\right] \qquad \text{and} \qquad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \\ b_{41} & b_{42} & b_{43} \\ b_{51} & b_{52} & b_{53} \end{bmatrix}.$$

How should B be partitioned to compute AB with block matrices.

Solution. Sol $\square$